

Phage Display

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A Genetically Encoded, Phage-Displayed Cyclic-Peptide Library

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Abstract: Superior to linear peptides in biological activities, cyclic peptides are considered to have great potential as therapeutic agents. To identify cyclic-peptide ligands for therapeutic targets, phage-displayed peptide libraries in which cyclization is achieved by the covalent conjugation of cysteines have been widely used. To resolve drawbacks related to cysteine conjugation, we have invented a phage-display technique in which its displayed peptides are cyclized through a proximity-driven Michael addition reaction between a cysteine and an amber-codon-encoded *N*^ε-acryloyl-lysine (AcrK). Using a randomized 6-mer library in which peptides were cyclized at two ends through a cysteine–AcrK linker, we demonstrated the successful selection of potent ligands for TEV protease and HDAC8. All selected cyclic peptide ligands showed 4- to 6-fold stronger affinity to their protein targets than their linear counterparts. We believe this approach will find broad applications in drug discovery.

Introduction

Traditionally, therapeutic drugs have consisted of small molecules that are exquisite at binding their receptors. However, due to their small size, small molecules have achieved little success at targeting proteins that use large, relatively flat surfaces for interactions with other molecules. With the development of the recombinant protein expression technology, a new class of protein pharmaceuticals, dubbed as biologics, has emerged. Because of their larger sizes, biologics display far superior target affinity and selectivity compared to small molecules. However, their increased size and protein-based

composition leads to poor tissue permeability and metabolic stability. With their intermediate size between small molecules and biologics, peptides offer a promising alternative to the two established classes of pharmaceuticals. Being larger than small molecules, peptides offer increased potency and target selectivity while maintaining a potential for cell permeability and a lower cost of manufacturing than biologics.^[1] Peptides are also extremely easy to screen. Using peptide-display technologies,^[2] such as phage display, which link the displayed peptide phenotype to the genotype, it is possible for a single researcher to screen a library of greater than 10¹⁰ unique peptides in a matter of days. Despite these advantages, peptide-based inhibitors have long been avoided for two reasons. First, peptides are generally unstructured in solution, which leads to an entropic penalty upon binding to a target. Second, peptides are highly susceptible to proteolysis when applied in vivo.^[3] It has long been known that macrocyclization can help to overcome some of the disadvantages of peptides.^[4,5] Macrocyclization imparts a degree of conformational rigidity to an unstructured peptide, which often increases the binding affinity of the peptide for its target.^[6] Cyclic peptides are also significantly more resistant to proteolysis.^[7] In several cases this has led to peptides so stable that they have been successfully used for oral delivery.^[8]

Although peptide cyclization generally leads to better pharmacological properties, cyclizing a linear peptide identified through screening can have unknown consequences on the ability of the peptide to bind to a target protein.^[9] For this reason, multiple binary coding systems have been developed for the direct selection of cyclic-peptide libraries.^[10] These include phage display, bacterial display, yeast display, traditional mRNA display, RaPID,^[4] and SICLOPPS,^[12] amongst others.^[11,12] Screening one-bead-one-compound libraries is also an effective method for the identification of potent cyclic-peptide ligands.^[13] Phages natively display linear peptides. To cyclize phage-displayed linear peptides, several chemical approaches have been developed.^[14] One involves the formation of a disulfide bond between two cysteine residues (Figure 1 A).^[11,15] There are many examples about using this strategy to produce disulfide-cyclized peptides with higher affinity for a target protein than their linear counterparts.^[16] While beneficial for some in vitro applications, peptides cyclized in this way cannot be used in vivo as they cannot withstand the reducing cellular environments. An alternative strategy relies on the reactivity of nucleophilic thiols towards small-molecule organic linkers to covalently connect two cysteines (Figure 1 B).^[17] This strategy has been successfully used for the formation of both mono- and bicyclic, phage-displayed peptide libraries and used to select ligands with

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Supporting information, including experimental details for the synthesis of all small molecules and peptides, protein expression, construction of plasmids, phagemids, and the phagemid library, selection, and characterization of selected peptides, and the ORCID identification number(s) for the author(s) of this article can be found under:
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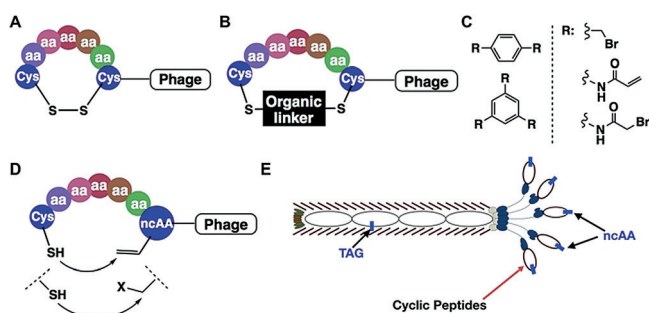


Figure 1. Representative, existing, and proposed cyclization strategies for phage-displayed peptides. A) Cyclization through disulfide-bond formation between cysteines. B) Cyclization through covalent conjugation of cysteines with organic linkers. C) Representative organic linkers used for cysteine conjugation to generate mono- and bicyclic peptides. D) A proposed proximity-driven cyclization between a cysteine and an electrophilic noncanonical amino acid (ncAA). E) An amber-codon-suppression-based approach to link the phenotypic ncAA with the genotypic TAG mutation. A phage with a TAG mutation at the coding region of its displayed peptide is produced in *E. coli* that harbour an evolved aminoacyl-tRNA synthetase and amber-suppressing tRNA for the genetic incorporation of the designated ncAA.

inhibition constants as low as 2 nM (Figure 1 C).^[18] Although effective at forming cyclic-peptide libraries, this method modifies native phage cysteines leading to low phage viability. Attempts have been made to construct phage strains with no surface cysteines.^[19] However, these phages have low viability, limiting the phage production. Due to the non-selective nature while conjugating cysteines, most current organic linkers are symmetric and achiral for avoiding heterogeneity in the phage-displayed cyclic peptides that might pose significant challenges in the following synthesis and characterization of selected cyclic peptides. The requirement of using symmetric, achiral organic linkers limits the structural diversity of cyclized peptides. To resolve these limitations, we envisioned that an electrophilic noncanonical amino acid (ncAA) and a cysteine can be genetically installed in phage-displayed peptides in close proximity for peptide cyclization (Figure 1 D). The incorporation of the ncAA into phages can be achieved by suppressing an amber mutation in the phage-displayed peptide coding region in *Escherichia coli* that harbor a ncAA-specific aminoacyl-tRNA synthetase–amber suppressor tRNA pair and grow in the presence of the ncAA (Figure 1 E).^[20] Using this method for the construction of a phage-display library will afford a genetically encoded, phage-displayed cyclic-peptide library whose spontaneous peptide cyclization requires neither the use of phage strains without surface cysteines nor an organic linker for cyclization. Encouragingly, several cases featuring proximity-driven reactions using a genetically encoded ncAA and a cysteine or another amino acid have recently been reported by Fasan, Suga, Wang, and their co-workers.^[21,22]

Results and Discussion

Pyrrolysine (Pyl) is the 22nd naturally occurring proteinaceous amino acid and is genetically encoded by an amber codon.^[23] Its incorporation is mediated by pyrrolysyl-tRNA

synthetase (PylRS) and tRNA. In the past decade, a number of research groups, including ours, have evolved PylRS for the genetic incorporation of more than 100 ncAAs including both lysine and phenylalanine derivatives into proteins in *E. coli*. One of these ncAAs is *N*^ε-acryloyl-lysine (AcrK), a Michael acceptor.^[24] We previously demonstrated that AcrK reacts slowly with a thiol (the second-order rate constant is 0.004 M⁻¹ s⁻¹) at physiological conditions but can be stably incorporated into proteins in *E. coli* using an evolved PylRS mutant (PrKRS) and tRNA^{Pyl}_{CUA}.^[24] The slow reaction between AcrK and cysteine is desirable in that it avoids non-specific reactions with cysteines in proteins but allows rapid conjugation when AcrK and cysteine are located in close proximity in a peptide. By installing a cysteine at one end and AcrK at the other end of a phage-displayed peptide, an automatic cyclization of the peptide is expected (Figure 2 A). There is also an advantage to work with AcrK. Its acrylamide moiety undergoes Huisgen 1,3-cycloaddition reaction selectively with a non-fluorescent diphenylnitrilimine moiety to form an intense blue fluorescent final product.^[25] Using HZC1 (Figure 2 B), which undergoes rapid dehydrochlorination in water to release a diphenylnitrilimine,^[26] protein or phage with intact AcrK can be easily labeled and visualized. However, a proximity-driven Michael-addition reaction between AcrK and cysteine in a peptide will annihilate the acrylamide moiety, leading to a cyclic peptide that cannot be labeled by HZC1.

To demonstrate the proximity-driven cyclization between a genetically incorporated AcrK and an adjacent cysteine in a protein, we expressed superfolder green fluorescent protein (sfGFP) that had a N-terminal CA₅X peptide (X denotes AcrK and is coded by an amber codon). To express this protein, we transformed *E. coli* BL21 (DE3) cells with two plasmids. One was a previously described pEVOL-PrKRS plasmid that contained both PrKRS and tRNA^{Pyl}_{CUA} genes^[24]

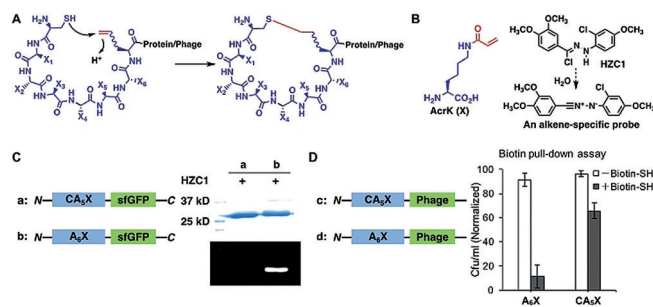


Figure 2. Cyclization of phage-displayed peptides through Michael addition between a cysteine and a genetically incorporated *N*^ε-acryloyl-lysine (AcrK). A) A diagram that illustrates proximity-driven peptide cyclization between a cysteine and an electron-deficient ncAA. B) The structure of AcrK and HZC1 whose dissociation product in water is a nitrilimine that reacts selectively with an acrylamide to show intense blue fluorescence. C) Two superfolder green fluorescent protein (sfGFP) derivatives, one with a N-terminal CA₅X peptide and the other with a N-terminal A₆X peptide, and their fluorescent labeling with HZC1. X denotes AcrK. Proteins were denatured and then analyzed by SDS-PAGE. In-gel fluorescence was recorded in the blue region with excitation at 365 nm. D) Two phage derivatives, one with a N-terminal CA₅X peptide and the other with a N-terminal A₆X peptide, and their titer results after reaction with the N-biotinyl-cysteamine probe and then capturing by streptavidin beads. Uncaptured phages were analyzed in the titer assay.

and the other was a pETDuet plasmid that contained a gene coding the CA₅X-sfGFP protein. Growing the transformed cells in the presence of AcrK afforded CA₅X-sfGFP. Labeling this protein with HZC1 did not produce a blue fluorescent product. However, a control sfGFP protein with an N-terminal A₆X peptide, which was expressed similarly to CA₅X-sfGFP and reacted with HZC1, provided an intense blue fluorescent protein band in a SDS-PAGE gel (Figure 2C). In parallel, we generated two phages, one with a CA₅X peptide and the other with an A₆X peptide at the N-terminus of the coating protein pIII. To construct the CA₅X phage, we inserted a CA₅X-coding DNA fragment between the PelB-leader-peptide-coding region and the phage-pIII-coding (gIII) gene in the pADLg3 phagemid vector, purchased from *Antibody Design Labs*. We used the afforded phagemid pADLg3-CA₅X to transform *E. coli* Top10 cells that also harbored a plasmid pEVOL-PrKRS-CloDF and a mutant helper phage plasmid M13K07-g3TAA. pEVOL-PrKRS-CloDF was derived from pEVOL-PrKRS by switching the origin of replication from p15a to CloDF for its compatible use with a helper phage plasmid that typically has a p15a origin of replication. We constructed M13K07-g3TAA by introducing a deleterious ochre mutation at the Q350 coding site in the gIII gene of the M13K07 helper phage. Since M13K07-g3TAA had a non-functional gIII gene, its use together with pADLg3-CA₅X drove the synthesis of a phage that contained pIII expressed only from pADLg3. Growing the transformed cells in the presence of AcrK afforded the CA₅X phage. We used the similar approach to produce the control A₆X phage. Following their separation, we incubated them together with N-biotinyl-cysteamine, a probe that reacts with acrylamide. We used 60 °C to speed up the relatively slow Michael-addition reaction. After the reaction, we used streptavidin beads to capture biotin-conjugated phages. Our result (Figure 2D) showed that more than 90 % A₆X phage was captured but about 65 % CA₅X phage remained in the supernatant, supporting that the A₆X phage had a free acrylamide moiety but the CA₅X phage underwent a proximity-driven cyclization to eliminate its acrylamide. The loss of phages during substantial washing to remove residual N-biotinyl-cysteamine might contribute to the observation that only about 65 % of the CA₅X phage was recovered. To examine whether peptide-chain lengths may significantly impact the proximity-driven cyclization between a cysteine and AcrK, we synthesized three Cys-containing peptides (CA₅X, CA₆X, and CA₇X) and their non-Cys control peptides (A₆X, A₇X, and A₈X). Labeling all six peptides with HZC1 in PBS buffer showed strong blue fluorescence for the non-Cys peptides but low fluorescence for the Cys-containing peptides, indicating that all Cys-containing peptides cyclized (Supporting Information, Figure S4). Increasing the peptide-chain length did not significantly affect the cyclization process.

Encouraged by the in vitro results, we advanced to construct a phage-displayed 6-mer cyclic-peptide library. To afford a phagemid library for the production of phages with displayed cyclic peptides, we inserted a 24 base-pair DNA fragment that encoded six randomized amino acids flanked by an N-terminal cysteine and a C-terminal AcrK between the

PelB-leader peptide-coding region and the gIII gene of the pADLg3 phagemid (Figure 3A). 20 clones from this library were sequenced to confirm the library diversity (Supporting Information, Figure S7). We used this phagemid library to transform *E. coli* Top10 cells that also contained pEVOL-PrKRS-CloDF and M13K07-g3TAA to afford close to 10⁹ transformants and then grew the transformed cells in the presence of AcrK to produce phages. To demonstrate the viability of using this library to select cyclic-peptide ligands for a protein target, we first tested it on a model protein, TEV protease, which was conjugated with biotin to enable its loading onto streptavidin magnetic beads for selection. We carried out three rounds of affinity-based selection. Eluted phages were clearly enriched after each round (Supporting Information, Figure S8). After the third round, we sequenced 25 phage clones that converged to only three peptide sequences, CycTev1, CycTev2, and CycTev3 (Supporting Information, Figure 9). Using solid-phase peptide synthesis, we synthesized 5-FAM-conjugated CycTev1, CycTev2, as well as their linear counterparts and then measured their binding affinities to TEV protease using fluorescence polarization assays. Our results as shown in Figure 3B–E and Table 1 indicate that both CycTev1 and CycTev2 bind to TEV protease with a single digit μ M dissociation constant and both cyclic peptides bind to TEV protease significantly better than their linear counterparts (greater than 6-fold). These results established the feasibility of using this genetically encoded phage-displayed cyclic peptide library to identify potent ligands for protein targets and demonstrated that cyclization contributes to the binding.

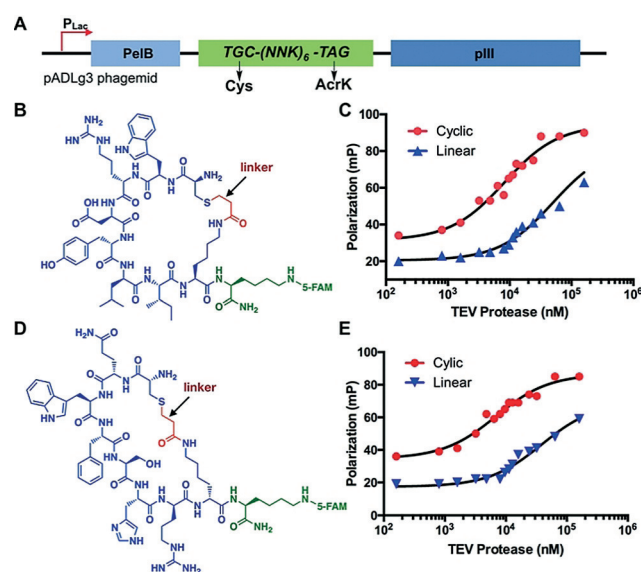


Figure 3. Selected TEV-protease-binding cyclic peptides and their K_d measurements. A) A diagram to show the phagemid structure for the production of a phage-displayed 6-mer cyclic-peptide library. B) The structure of 5FAM-CycTev1. CycTev1, highlighted in blue and red, was selected from phage display. C) Fluorescence polarization analysis of 5FAM-CycTev1 binding to TEV protease. Data for a linear counterpart of 5FAM-CycTev1 with no linker is also included. D) The structure of 5FAM-CycTev2; E) Fluorescence polarization analysis of 5FAM-CycTev2 binding to TEV protease. Data for a linear counterpart of 5FAM-CycTev2 with no linker is also included.

Tabelle 1: Determined K_d and IC_{50} values of selected cyclic peptides and their linear counterparts when binding to their protein targets.

Ligand	Sequence ^[a]	Protein target	K_d [μ M]	IC_{50} [μ M]
CycTev1	CWRDYLYX	TEV protease	8.2 ± 0.8	
LinTev1	CWRDYLIK	TEV protease	50 ± 5	
CycTev2	CQWFSHRX	TEV protease	6.9 ± 0.9	
LinTev2	CQWFSHRK	TEV protease	39 ± 5	
CycH8a	CQSLWMNX	HDAC8	7.1 ± 0.7	9.7 ± 0.7
LinH8a	CQSLWMNK	HDAC8	> 50	
LinH8a'	CQSLWMNNle	HDAC8	31 ± 7	

[a] X denotes AcrK and Nle denotes norleucine.

HDAC8 is a Zn^{2+} -dependent histone deacetylase that has been implicated as a therapeutic target in various diseases, including cancer, X-linked intellectual disability, and parasitic infections.^[27] Notable efforts have been made to identify potent HDAC8 inhibitors.^[28] To identify novel cyclic-peptide ligands for HDAC8, we carried out selection from our genetically encoded, phage-displayed 6-mer cyclic-peptide library similar to that for TEV protease. Out of selected clones that we subsequently sequenced, the majority converged to a single sequence CycH8a (Table 1 and Supporting Information, Figure S10).

To determine the affinity of CycH8a for HDAC8, we synthesized 5-FAM-conjugated CycH8a (Figure 4A) and then characterized its binding to HDAC8 using fluorescence polarization analysis. The result indicated a 7.1μ M dissociation constant (Figure 4B). We also synthesized 5-FAM-conjugated LinH8a, a linear counterpart of CycH8a, and tested its binding to HDAC8. However, LinH8a bound very weakly to HDAC8. Due to the fact that HDAC8 aggregated at a concentration higher than 100μ M, we were not able to collect enough data to accurately determine K_d for LinH8a, though it is estimated to be higher than 50μ M. LinH8a has a positively charged lysine residue; however, CycH8a has this

charge neutralized. To resolve a concern that this charge difference might contribute to the binding disparity between the two peptides, we synthesized 5-FAM-conjugated LinH8a, in which norleucine (Nle), a neutral amino acid spatially similar to lysine was installed at the lysine position in LinH8a and determined its dissociation constant toward HDAC8 as 31μ M (Table 1 and Supporting Information, Figure S12). This K_d value is four-fold higher than that of CycH8a, eliminating the possibility that the neutral charge of CycH8a contributes significantly to its strong binding to HDAC8. Therefore, the cyclization is critical to providing high potency to CycH8a for its binding to HDAC8. For a protein target, it does not necessarily bind to the active site of the protein for direct inhibition. To test whether CycH8a can directly inhibit the deacetylation activity of HDAC8, we adopted a HDAC8 activity assay^[29] as shown in Figure 4C and synthesized the substrate Boc-AcK-AMC. In this assay, HDAC8 catalyzes the deacetylation of Boc-AcK-AMC to afford Boc-K-AMC that reacts with the coupling enzyme trypsin to release the fluorescent AMC, a compound that we could easily track in a fluorescent plate reader. As shown in Figure 4D, providing CycH8a to the assay inhibited the deacetylation of Boc-AcK-AMC by HDAC8. The determined IC_{50} value in the conditions of 5μ M HDAC8 and 50μ M Boc-AcK-AMC is 9.7μ M, close to the determined K_d value. Given that IC_{50} is not a direct binding-affinity indicator and influenced by the concentration of the substrate, its slightly higher value than K_d was expected. Collectively, the results in Figure 4 demonstrated a successful application of using this genetically encoded, phage-displayed cyclic-peptide library in identifying a potent cyclic-peptide inhibitor for a therapeutic protein target.

To gain insight into how CycH8a might interact with HDAC8, we virtually docked CycH8a on HDAC8. HDAC8 naturally occurs in a dimeric form.^[30] Therefore, we investigated both monomeric and dimeric HDAC8 as the receptor for docking. The docking results indicated that CycH8a binds weakly to monomeric HDAC8 but fits favorably in two grooves that are at the dimeric interface of HDAC8 and close to the two active sites (Figure 5). Several published crystal structures of HDAC8–substrate complexes have shown that the two dimer-interface grooves are also part of channels required for binding peptide substrates. The binding of CycH8a at the two grooves will block the entry of a peptide substrate into the two active sites, which provides an explanation for the inhibition of HDAC8 by CycH8a. This is different from most HDAC8 inhibitors that have been developed so far. Most HDAC8 inhibitors are hydroxamates or other metal binders that strongly chelate the active site Zn^{2+} . They bind strongly to the active site tunnel of HDAC8 with IC_{50} values typically below 1μ M.^[31] Our modeling results show that CycH8a does not directly interact with Zn^{2+} and other residues in the active site, explaining its relatively lower potency than hydroxamate inhibitors. However, CycH8a binds close to the active-site tunnel of HDAC8. One possibility to develop more potent HDAC8 inhibitors is to conjugate an active-site-targeting hydroxamate inhibitor and CycH8a to form a tight-binding, bidentate ligand. This is a research direction that we are actively exploring.

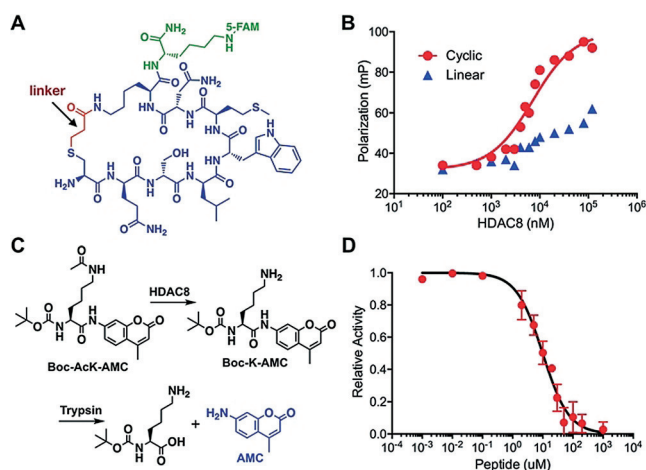


Figure 4. A selected cyclic-peptide ligand, CycH8a, and its binding and inhibition of HDAC8. A) The structure of 5FAM–CycH8a. CycH8a, highlighted in blue and red, was selected from phage display. B) Fluorescence polarization analysis of 5FAM–CycH8a binding to HDAC8. Data for a linear counterpart of 5FAM–CycH8a is also included. C) A diagram to show a fluorogenic HDAC8 activity assay Scheme. D) The IC_{50} determination of 5FAM–CycH8a inhibition of HDAC8 using the assay shown in (C).

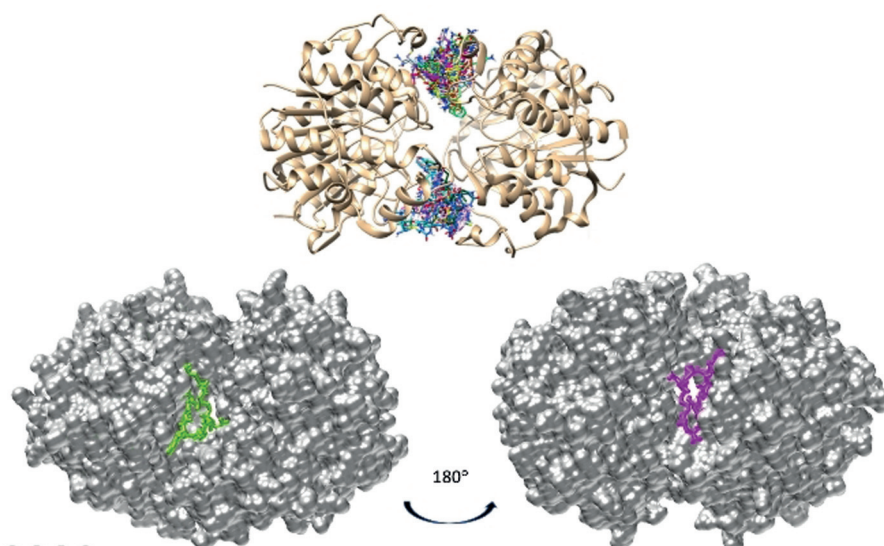


Figure 5. The molecular docking results of Cych8a binding to the HDAC8 dimer. The top panel shows different Cych8a conformers binding at two grooves of the HDAC8 dimer interface. The bottom panel presents the most favorable conformer of Cych8a binding at each groove.

Conclusion

In summary, we have developed a phage-display technique that allows the construction of a genetically encoded, phage-displayed cyclic-peptide library. The cyclization of phage-displayed peptides is achieved by a proximity-driven Michael-addition reaction between a cysteine and an AcrK that flank a randomized 6-mer peptide sequence. AcrK was encoded by an amber codon and its incorporation into phages was mediated by an evolved PylRS-tRNA^{Pyl}_{CUA} pair in *E. coli*. Applying the developed library to selection against both TEV protease and HDAC8 afforded cyclic-peptide ligands that bind to their protein targets with single digit μM K_d values and significantly better than their linear counterparts. As a proof of concept, the current study involved relatively small peptides that randomized only 6 residues. It is expected that a library with much bigger randomized peptides will afford the selection of more potent ligands. Given that many electrophilic ncAAs have been incorporated into proteins using the amber-suppression-mutagenesis approach,^[22,32] they can all potentially be used to construct genetically encoded, phage-displayed cyclic-peptide libraries. Since these ncAAs are structurally diverse, their use will impart different structural constraints to phage-displayed cyclic peptides that will provide diverse structural diversity, which is beneficial for selection. Unlike most other binary coding techniques for the construction of cyclic-peptide libraries, the reported method leads to direct, irreversible, and simultaneous cyclization of displayed peptides right after their translation. There is no additional chemical intervention required. This feature, shared by the mRNA-display-based RAPID technique,^[4] significantly simplifies the construction of cyclic-peptide libraries. As an addition to the phage-display technique, we anticipate that this technique will find broad applications in the identification of potent ligands for many surface receptors and strong inhibitors for enzymes and protein–protein/DNA/RNA interactions.

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Conflict of interest

The authors declare no conflict of interest.

Stichwörter: Cyclische Peptide · HDAC8 · N^ε-Acryloyllysine · Phagendisplay · Proximitätsgetriebene Cyclisierung

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